CA2 Overview (49%)

This document describes the CA2 requirements for the EE4409. The group project is a larger assignment where you will work in teams to analyze one of the domains in more depth and propose data-driven improvements. Each group will consist of 5 members. Each team will select exactly one of the three domains: Building Energy, Manufacturing Plant, or Plantation Monitoring.

Insights and limitations identified in the CA1 may be used to motivate your system improvements and design choices.

Project deliverables:

1. A written project report (8–10 pages). [14%]
2. A dashboard or set of visualizations based on the dataset. [7%]
3. An in-class presentation summarizing your work and recommendations [Max time would be 15 mins per group]. [21 %]
4. Peer assessment [Evaluating the strengths and weaknesses of **at least 7 other groups**]. [7%]

<u>The explanations must be written in your own words.</u>

General expectations for CA2

For your chosen domain, your project should:

- Clearly describe the real-world system being monitored and the problem or objective (for example, reducing energy wastage or improving crop health).
- Explain the sensors used and what each of them measures, including units and typical ranges.
- Analyze the dataset to understand typical behavior, trends over time, and differences between zones/machines/plots.
- Identify anomalies, inefficiencies, or risk conditions using data.
- Design and present a new dashboard that would help a non-expert user monitor the system.
- You are not limited to the sensor types present in the dataset and may propose alternative or additional sensing modalities if they better address the monitoring objective.
- Provide clear, practical recommendations based on your analysis.
- Clearly justify why the proposed sensing approach is appropriate compared to existing measurements, considering practicality, cost, and information gained.
- Include one-page executive summary and state explicitly the contributions of each team members

You may use programming languages, spreadsheet tools, BI tools, and AI assistants to support your work. However, during presentations or questioning, students may be asked to explain or justify any part of the analysis or dashboard.

<u>A suggested format for report would be problem framing (10%), system & sensor design understanding (30%), observations and patterns (15%), interpretation & reasoning (30%), and dashboard design (15%).</u>
This breakdown is indicative; emphasis will be placed on clarity of reasoning and system-level justification.

Guidelines on Use of AI Tools

You are encouraged to use AI assistants and other digital tools to help with:

- Dashboard design.
- Creating plots, tables, and visualizations.
- Brainstorming interpretations and recommendations.

However, you are expected to:

- Understand every step of your analysis and be able to explain it.
- Check AI outputs for correctness, especially units, ranges, and formulas.

Guidelines on Presentation

The presentation should focus on key insights, system understanding, and recommendations rather than detailed methodology.

- Presentation duration would be 15 minutes for each group and 3 minutes of Q&A.
- During presentations, students may introduce their designed dashboard briefly.

Guidelines on Peer Assessment

You are encouraged to observe the work from your peers and reflect on the following aspects for at least 5 other groups.

- One strong insight you learned
- One sensing/design limitation
- One suggestion for improvement

Submission Deadline

The report's submission must be done by 3rd April. Late submissions will incur penalties (10% deduction for a day).

Presentations will be on the 12th and 13th week (10th and 17th April). We will inform the finalized schedules by 3rd April.